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9702/52

May/June 2011

Candidates answer on the Question Paper.

No Additional Materials are required.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
Total	

This document consists of **8** printed pages.

- 1 A student wishes to investigate projectile motion.

A small ball is rolled with velocity v along a horizontal surface. When the ball reaches the end of the horizontal surface, it falls and lands on a lower horizontal surface. The vertical displacement of the ball is p and the horizontal displacement of the ball is q , as shown in Fig 1.1.

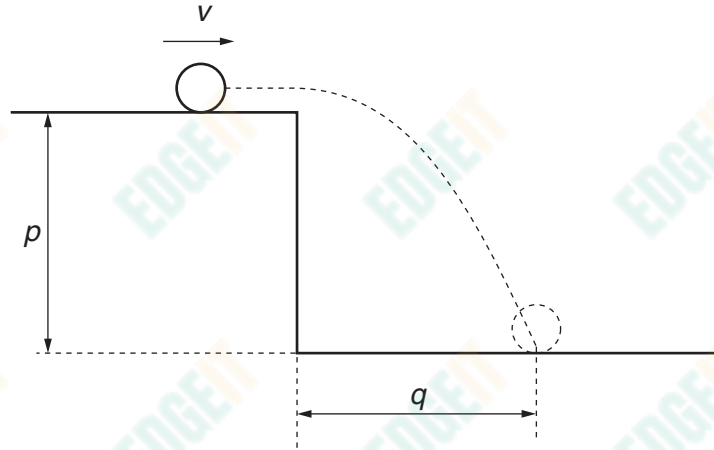


Fig. 1.1

It is suggested that

$$gq^2 = 2pv^2$$

where g is the acceleration of free fall.

Design a laboratory experiment to investigate how q is related to p and how v may be determined from the results. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- (a) the procedure to be followed,
- (b) the measurements to be taken,
- (c) the control of variables,
- (d) the analysis of the data,
- (e) the safety precautions to be taken.

[15]

Diagram

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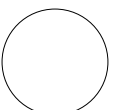
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This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. A faint, repeating diagonal watermark is visible across the entire surface, consisting of the word "EDGE" in a stylized font, colored in shades of blue and orange. The watermark is oriented diagonally from the bottom-left towards the top-right. There are no margins, text, or other markings on the page.

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- 2 A student is investigating a non-inverting operational amplifier (op-amp) circuit.

The circuit is set up as shown in Fig. 2.1.

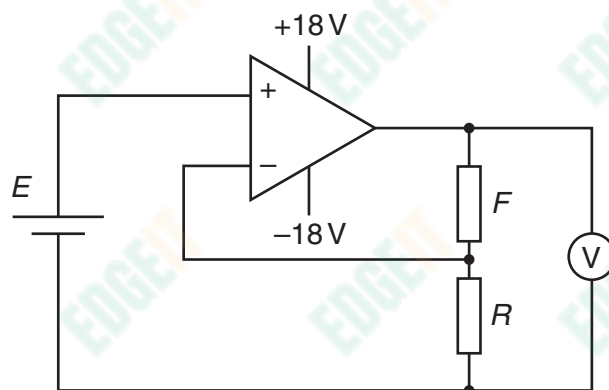


Fig. 2.1

The op-amp is connected to a +18V and -18V power supply.

E is the e.m.f. of the cell, which has a value of 1.6 ± 0.1 V.

An experiment is carried out to investigate how the reading V on the voltmeter varies with resistance R .

Question 2 continues on the next page.

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It is suggested that V and R are related by the equation

$$V = \frac{F}{R} E + E$$

where F is the resistance of the fixed resistor in the circuit.

- (a) A graph is plotted of $\frac{V}{E}$ on the y-axis against $\frac{1}{R}$ on the x-axis. Express the gradient in terms of F .

gradient =[1]

- (b) Values of R and V are given in Fig. 2.2.

R/Ω	V/V	$\frac{1}{R}/10^{-3}\Omega^{-1}$	$\frac{V}{E}$
150	14.4 ± 0.1		
220	10.4 ± 0.1		
330	7.4 ± 0.1		
470	5.6 ± 0.1		
680	4.4 ± 0.1		
860	3.8 ± 0.1		

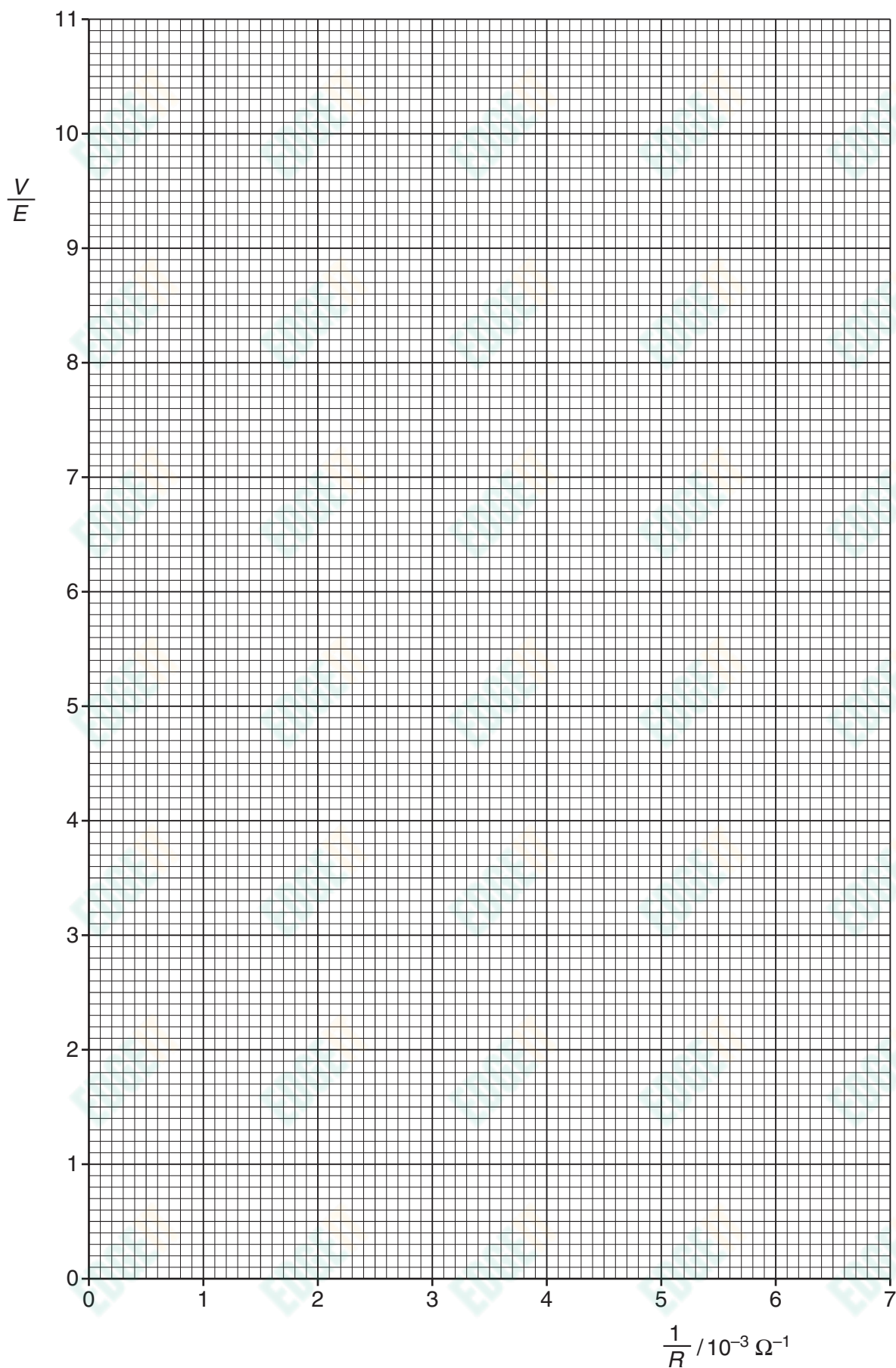
Fig. 2.2

Calculate and record values of $\frac{1}{R}/10^{-3}\Omega^{-1}$ and $\frac{V}{E}$ in Fig. 2.2. Include the absolute uncertainties in $\frac{V}{E}$. [3]

- (c) (i) Plot a graph of $\frac{V}{E}$ against $\frac{1}{R}/10^{-3}\Omega^{-1}$. Include error bars for $\frac{V}{E}$. [2]
- (ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Both lines should be clearly labelled. [2]
- (iii) Determine the gradient of the line of best fit. Include the uncertainty in your answer.

gradient =[2]

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- (d) Using your answer in (c)(iii), determine the value of F . Include the absolute uncertainty in your value and an appropriate unit.

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$$F = \dots\dots\dots [2]$$

- (e) For one measurement, R has a value of $120\,\Omega \pm 5\%$.

- (i) Determine the value of $\frac{V}{E}$ using the relationship given and your answer in (d).
Include the absolute uncertainty in your answer.

$$\frac{V}{E} = \dots\dots\dots [2]$$

- (ii) Determine the expected voltmeter reading.

$$\text{voltmeter reading} = \dots\dots\dots \text{ V } [1]$$

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